

TE Curve Verification Pipeline Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Verdict

This update accepts the completed `wave 3.2` harmonic-offset probe into the official `TE Curve Verification Pipeline` offline verification matrix.

Decision

- `wave 3.2` is verified as an exploratory harmonic-offset baseline.
- No `wave 3.2` candidate is promoted over the current accepted `TE Curve Verification Pipeline` leaders.
- The project continues to maintain three parallel best surfaces: `Fw` , `Bw` , and `global` .
- Within `wave 3.2` , the strongest forward candidate is `track2f_bis_harmonic_residual_offset_Fw` .
- Within `wave 3.2` , the strongest backward candidate is `track2f_bis_harmonic_residual_offset_Bw` .
- Within `wave 3.2` , the strongest global candidate is `track2f_bis_clean_sequential_residual_offset_global` .

Source Package

This official report consolidates these refreshed artifacts:

- metric matrix:
doc/reports/analysis/track2/Track 2 Directional Model Comparison.md ;
- matrix summary:
output/validation_checks/track2_reference_comparison/2026-06-08-13-25-37__track2_full_directional_family_matrix_track2f_bis_harmonic_offset_probe_track2_refresh_2026_06_08/validation_summary.yaml ;
- per-condition metrics:
output/validation_checks/track2_reference_comparison/2026-06-08-13-25-37__track2_full_directional_family_matrix_track2f_bis_harmonic_offset_probe_track2_refresh_2026_06_08/per_condition_metrics.csv ;
- best-model collage report:
doc/reports/analysis/track2/best_model_collage_report/[2026-06-08]/track2_best_model_collage_report.md ;
- multi-model curve comparison report:
doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-06-08]/track2_multi_model_curve_comparison_report.md .

Candidate Refresh

The refresh added 6 registry-backed Wave 3.2 candidates and rechecked the 3 existing Wave 3.1 candidates in the same incremental package.

Surface	Branch	Candidate	Registry
global	clean sequential residual offset	track2f_bis_clean_sequential_residual_offset_global	output/registries/families/track2f_bis_clean_sequential_residual_offset_global/latest_family_best.yaml
Fw	clean sequential residual offset	track2f_bis_clean_sequential_residual_offset_Fw	output/registries/families/track2f_bis_clean_sequential_residual_offset_fw/latest_family_best.yaml
Bw	clean sequential residual offset	track2f_bis_clean_sequential_residual_offset_Bw	output/registries/families/track2f_bis_clean_sequential_residual_offset_bw/latest_family_best.yaml
global	harmonic residual offset	track2f_bis_harmonic_residual_offset_global	output/registries/families/track2f_bis_harmonic_residual_offset_global/latest_family_best.yaml
Fw	harmonic residual offset	track2f_bis_harmonic_residual_offset_Fw	output/registries/families/track2f_bis_harmonic_residual_offset_fw/latest_family_best.yaml
Bw	harmonic residual offset	track2f_bis_harmonic_residual_offset_Bw	output/registries/families/track2f_bis_harmonic_residual_offset_bw/latest_family_best.yaml

The matrix now contains 120 candidates. The incremental operator run used the completed Wave 3.1 refresh as its 114 -candidate baseline and evaluated 9 registry-backed offset-aware candidates.

Verification Rules

Surface	Training or Archive Scope	Evaluation Scope
Fw	forward-only training or archive	forward TE Curve Verification Pipeline curves only
Bw	backward-only training or archive	backward TE Curve Verification Pipeline curves only
global	forward and backward training together	both directions, reported by direction and combined

The `global` , `Fw` , and `Bw` branches are carried forward in parallel. They are not collapsed into one scalar winner.

Current Leaders

Best Composite References

Candidate	Source	Direction	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
paper_retuned_best_Fw	rcim_retuned	forward	0.001839	0.002041	4.109	9.866
paper_retuned_best_Bw	rcim_retuned	backward	0.003675	0.004284	7.572	15.645

Repository-Owned Static Baselines

Direction	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
forward	tree_global	0.002998	0.003364	6.590	11.601
backward	tree_Bw	0.003258	0.003651	7.051	14.116
global combined	tree_global	0.003144	0.003533	6.854	13.314

Repository-Owned Neural Leaders

Surface	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
Fw	periodic_gru_sequence_Fw	0.003186	0.003438	7.077	13.323
Bw	periodic_gru_sequence_Bw	0.002392	0.002639	5.466	14.820
global combined	periodic_gru_sequence_global	0.002704	0.002949	6.139	13.396

Wave 3.2 Result

Surface	Candidate	Branch	MAE [deg]	RMSE [deg]	Mean [%]
Fw	track2f_bis_harmonic_residual_offset_Fw	harmonic residual offset	0.002850	0.003108	6.286
Fw	track2f_bis_clean_sequential_residual_offset_Fw	clean sequential residual offset	0.003439	0.003870	7.632
Bw	track2f_bis_harmonic_residual_offset_Bw	harmonic residual offset	0.003331	0.003671	7.261
Bw	track2f_bis_clean_sequential_residual_offset_Bw	clean sequential residual offset	0.003541	0.003971	7.732
global forward slice	track2f_bis_harmonic_residual_offset_global	harmonic residual offset	0.003255	0.003547	7.224

Surface	Candidate	Branch	MAE [deg]	RMSE [deg]	Mean [%]
global backward slice	track2f_bis_harmonic_residual_offset_global	harmonic residual offset	0.003805	0.004120	8.354
global combined	track2f_bis_clean_sequential_residual_offset_global	clean sequential residual offset	0.003522	0.003950	7.754
global combined	track2f_bis_harmonic_residual_offset_global	harmonic residual offset	0.003530	0.003833	7.789

The harmonic branch materially improves the direction-specific `Fw` result and also improves the direction-specific `Bw` result versus the clean sequential residual offset control. The same conclusion does not hold for the `global` surface: the harmonic global model improves its forward slice, but degrades the backward slice enough that the clean global control remains the stronger Wave 3.2 global candidate by combined MAE.

Visual Evidence

The `2026-06-08` collage and overlay bundles were regenerated with dedicated Wave 3.2 sections:

- `Forward Wave 3.2 Harmonic-Offset Probe Models` ;
- `Backward Wave 3.2 Harmonic-Offset Probe Models` ;
- `Global Wave 3.2 Harmonic-Offset Probe Models` in the collage report.

The visual package confirms the same decision pattern as the matrix: harmonic forcing is useful for the direction-specific branches, while the global branch still needs a better way to balance forward and backward offset/shape behavior.

Campaign Update Ledger

Date	Campaign or Update	Candidate Scope	Matrix Status	Visual Status	Decision
2026-06-08	Wave 3.2 harmonic-offset probe refresh	6 clean and harmonic <code>global</code> , <code>Fw</code> , and <code>Bw</code> candidates plus 3 rechecked Wave 3.1 candidates	included in the 120 - candidate matrix	dated collage and overlay bundles regenerated with Wave 3.2 sections	verified exploratory baseline; not promoted
2026-06-04	Wave 3.1 offset-aware probe refresh	3 <code>global</code> , <code>Fw</code> , and <code>Bw</code> sequential residual offset candidates	included in the 114 - candidate matrix	dated collage and overlay bundles regenerated	verified exploratory baseline; not promoted
2026-05-28	Wave 2.3 residual harmonic temporal hybrid refresh	18 <code>global</code> , <code>Fw</code> , and <code>Bw</code> residual harmonic GRU/LSTM candidates	included in the 111 - candidate matrix	collage and overlay reports refreshed with Wave 2.3 sections	verified exploratory baseline; not promoted over Wave 2.2 or accepted TE Curve Verification Pipeline baselines
2026-05-26	Wave 2.2 harmonic temporal hybrid refresh	periodic temporal convolution, GRU, and LSTM <code>global</code> , <code>Fw</code> , and <code>Bw</code> candidates	included	collage and overlay reports refreshed	strongest repository-owned neural branch

Closeout Decision

Wave 3.2 does not change the accepted TE Curve Verification Pipeline baseline. The current direction-parallel decision remains:

- Fw : rcim_retuned_GBM19_Fw remains the strongest overall forward candidate, while track2f_bis_harmonic_residual_offset_Fw is a useful harmonic-offset exploratory branch.
- Bw : periodic_gru_sequence_Bw remains the strongest practical repository-owned backward candidate; track2f_bis_harmonic_residual_offset_Bw is better than the clean offset control but is not promoted.
- global : periodic_gru_sequence_global remains the strongest repository-owned bidirectional neural candidate; within Wave 3.2, the clean global control is slightly stronger than the harmonic global branch. The next modeling step should keep harmonic forcing available for direction-specific probes, but should not treat it as sufficient for the global surface. The next planned branch should move to a curve-aware loss or a multi-task shape/offset structure that can balance raw error, offset, centered shape, amplitude, and phase while preserving the causal runtime input contract.